

LAPTOP PRICE SPECIFICATION ANALYSIS

Project Report | Python Data Analysis and Power BI Dashboard

Project domain: E-commerce / Consumer Electronics

Project focus: Understanding how laptop brands, hardware specifications, laptop type, storage, operating system, screen size and weight relate to product pricing.

1. Executive Summary

This project analyses a laptop-price dataset and presents the findings through Python-based data preparation and exploratory data analysis (EDA), followed by a multi-page Power BI dashboard. The notebook documents a dataset of 1,370 rows and 13 original columns, covering laptop identifiers, brands, product types, screen size, resolution, processor, RAM, storage, graphics, operating system, weight and price.

The purpose is to turn raw product records into a more useful analytical dataset, compare prices across brands and laptop types, study specification patterns, and communicate the results through charts, KPI cards, tables and slicers.

Project objectives

- Identify which product characteristics appear alongside higher or lower laptop prices.
- Compare brand representation and average price across laptop companies.
- Explore the role of RAM, processor, storage, operating system, screen size, weight and laptop type.
- Prepare derived fields to support comparison, grouping and dashboard filtering.
- Present practical findings for customer product comparison and business inventory/pricing decisions.

Project workflow

1. Define the problem and select the laptop dataset.
2. Inspect and clean the data in Python using pandas and NumPy.
3. Create visualisations and grouped analyses with Matplotlib and Seaborn.
4. Export the cleaned/engineered data to CSV for reporting.
5. Build a Power BI dashboard with summary KPIs, comparisons, tables and slicers.

2. Dataset and Python Environment

Dataset overview

Item	Details recorded in the notebook
Dataset name	Laptop Price Specification Analysis
Source / file	laptop_price.csv (notebook notes GitHub as source)
Original size	1,370 rows × 13 columns
Period noted	2024–2025 in the notebook description
Domain	E-commerce / Consumer Electronics
Target area	Laptop price and specification comparison

Original fields

The notebook lists these original fields: laptop_ID, Company, Product, TypeName, Inches, ScreenResolution, Cpu, Ram, Memory, Gpu, OpSys, Weight and Price_euros. Together they describe product identity, company, device category, screen, hardware, operating system, physical weight and price.

Libraries and loading

- pandas (pd): loading CSV files, inspecting columns, handling missing values, renaming fields, grouping and pivot tables.
- numpy (np): numerical operations and the logarithmic price transformation.
- Matplotlib (plt): figure sizing, chart titles, axes and rendering plots.
- Seaborn (sns): count plots, histograms, boxplots, scatterplots, heatmaps and pairplots.
- warnings: suppresses warning messages in notebook output.

The data is loaded with `pd.read_csv("laptop_price.csv", encoding="latin-1")`. The notebook then uses `head()`, `tail()`, `shape`, `info()`, `columns.tolist()`, `dtypes` and `describe()` to inspect sample records, dimensions, field types and numerical summaries.

Why this step matters

Initial inspection establishes what the dataset contains before changes are made. It also helps identify which fields are categorical, which should be numeric, and which variables may be useful for price comparisons.

3. Data Cleaning and Pre-processing in Python

Missing values and duplicates

The notebook checks missing values with `df.isnull().sum()` and removes rows with missing values using `df.dropna()`. It also checks repeated records with `df.duplicated().sum()`; the notebook notes that the duplicate check returned zero. Weight is stripped of the “kg” unit and converted to numeric values; missing numeric weights are filled with the mean. RAM is stripped of “GB”, converted to numeric, and missing values are filled with the median.

The notebook also fills missing price values with the mode. These steps aim to make the relevant fields consistent and usable for analysis. Because the notebook performs `dropna()` before later imputation, the order and actual missing-value counts should be rechecked if the source data changes.

Column renaming and removal

Price_euros is renamed to price, and TypeName is renamed to Type. Product and ScreenResolution are dropped because the notebook marks them as not required for the later analysis. This simplifies the working table, while retaining the fields used in charts and dashboard comparisons.

Outliers and skewness

The notebook applies the interquartile range (IQR) rule to the price field. It calculates the first quartile (Q1), third quartile (Q3), and $IQR = Q3 - Q1$, then filters out price values outside $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$. It also calculates price skewness and creates price_log with `np.log1p(df["price"])`.

The IQR method can reduce the influence of unusually high or low values, but removing outliers may also exclude legitimate premium laptop models. The filtered row count should be recorded so the impact is transparent.

Data type checks

The notebook checks the data types of Ram, Weight and Inches. Converting these fields into numeric values allows averages, distributions, correlations and grouped calculations to work correctly.

Result of the cleaning stage

The result is a more analysis-ready dataset with consistent numeric fields, clearer column names, selected columns removed, and a price outlier filter. The notebook exports the resulting dataframe as `laptop_price_Cleaned.csv`.

4. Feature Engineering and Exploratory Data Analysis

New and transformed fields

Field	How it is created / purpose
Weight_per_Inch	Weight ÷ Inches; a simple measure for comparing weight relative to screen size.
Ram_category	Bins RAM into LOW, MEDIUM, HIGH, VERY HIGH and EXTREME HIGH groups.
discounted_price	price × 0.90, representing a hypothetical 10% discount.
RAM_GB	Numeric copy of Ram for analysis and visuals.
Screen_size	Numeric copy of Inches for charting.
Weight_kg	Numeric copy of Weight for charting.
price_inr	price × 100 in the current notebook code; the multiplier should be verified before calling this a true currency conversion.
final_price	price_inr – discounted_price in the current notebook code; see validation note below.
price_log	np.log1p(price), used to reduce the effect of large values/skewness.

Important calculation validation

The notebook currently calculates $\text{price_inr} = \text{price} \times 100$ and $\text{final_price} = \text{price_inr} - \text{discounted_price}$. The notebook screenshots use “final price” values in the dashboard, but the code does not document an exchange-rate source for the $\times 100$ multiplier, and subtracting `discounted_price` from `price_inr` mixes values derived from different scales. Before submitting the project as an Indian-rupee price analysis, confirm the intended conversion rate and define `final_price` consistently (for example, price converted to INR, or price after discount—not a subtraction between unlike units). The report therefore describes these as the formulas present in the notebook, not as a verified currency conversion.

Visualisations in the notebook

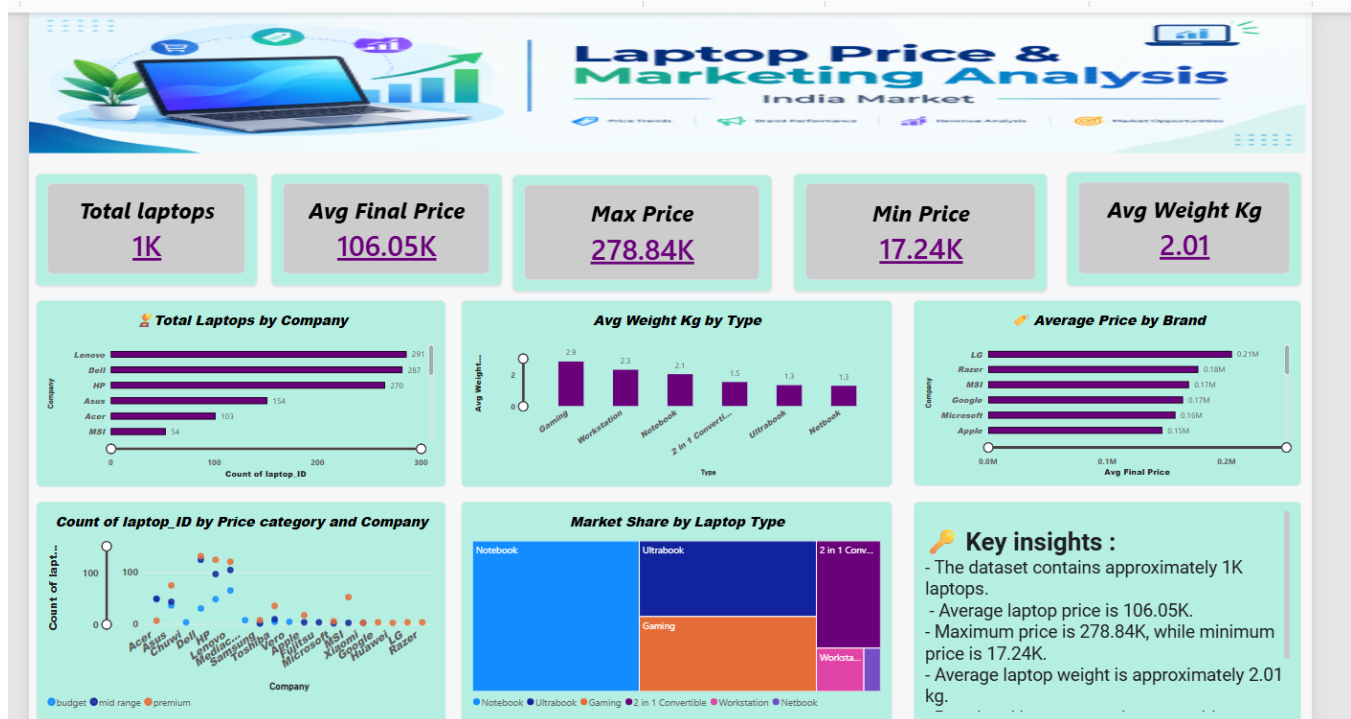
- Company-wise count plot: compares how many laptop records are present for each company.
- Discounted-price histogram with KDE: shows the distribution and frequency of hypothetical discounted prices.
- Weight boxplot: shows the median, middle 50% spread and potential weight outliers.
- Type vs final-price scatter plot, coloured by company: explores price variation across laptop types and brands.
- Average screen size by laptop type bar chart: compares screen dimensions between categories.
- RAM vs screen-size correlation heatmap: reports a weak positive correlation of about 0.19 in the notebook.
- Pairplot using RAM_GB, Screen_size and Weight_kg, with OpSys as hue: compares numerical relationships and distributions.
- Grouped analysis of RAM category, company and Weight_per_Inch: compares selected brand/RAM combinations.
- Grouped line chart of average weight and screen size by laptop type: intended to examine portability and display-size patterns.
- Pivot-table heatmap of company × type using discounted_price sums: compares value across company/type combinations.
- Grouped bar plot of company × laptop type against final_price: compares the price measure by brand and product type.

5. Power BI Dashboard – Pages and Visuals

The supplied screenshots show a three-page Power BI dashboard. The pages use a consistent mint/teal canvas, dark titles, purple/blue/orange chart colours, KPI cards, chart panels and key-insight text areas.

Page 1: Laptop Price & Marketing Analysis – India Market

- Top KPI cards: Total Laptops (about 1K), Avg Final Price (106.05K), Max Price (278.84K), Min Price (17.24K) and Avg Weight (2.01 kg).
- Total Laptops by Company: horizontal bar chart showing company representation; Lenovo, Dell and HP appear among the largest groups in the displayed chart.
- Average Weight by Type: compares average weight across Gaming, Workstation, Notebook, 2-in-1 Convertible, Ultrabook and Netbook categories.
- Average Price by Brand: ranks brands by the displayed average final-price measure.
- Count of laptop_ID by Price Category and Company: scatter plot compares budget, mid-range and premium groupings across brands.
- Market Share by Laptop Type: treemap displays the relative representation of laptop types.
- Key insights panel: summarises dataset size, price range and average weight.



Page 2: Price Distribution & Product Comparison

- KPI cards: Avg_price_SSD (130.45K), Avg_price_HDD (93.82K) and Sum of Storage_GB (566.74K), as displayed in the screenshot.
- Price vs RAM Size: compares RAM categories with the displayed price and RAM measures.
- Storage Type vs storage capacity: treemap compares capacity categories.
- GPU & CPU Brand Performance Analysis: detailed table for processor-related records and HIGH/LOW/MEDIUM groupings.
- Average Price by Operating System: horizontal bar chart compares macOS, Windows versions, Linux, No OS, Chrome OS and Android as labelled in the visual.
- Key insights panel: explains the roles of the treemap, column chart, detail table and horizontal bar chart.

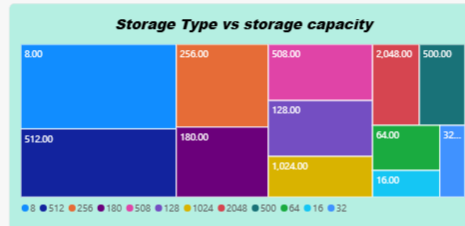
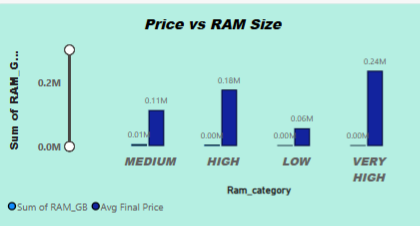


Price Distribution & Product Comparison

Avg_price_SSD
130.45K

Avg_price_HDD
93.82K

Sum of Storage_GB
566.74K

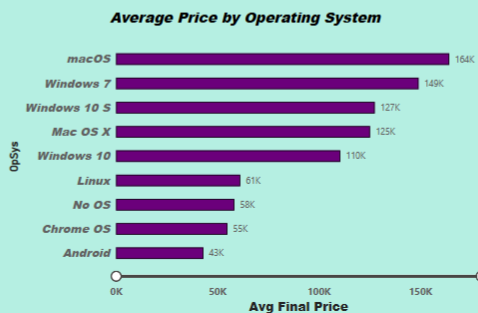


GPU & CPU Brand Performance Analysis

Cpu	HIGH	LOW	MEDIUM
Samsung Cortex A72&A53 2.0GHz		65,306.90	
Intel Xeon E3-1505M V6 3GHz			1,97,506.30
Intel Pentium Quad Core N4200 1.1GHz		42,548.09	
Intel Pentium Quad Core N3710 1.6GHz		41,225.49	38,549.90
Intel Pentium Quad Core N3700 1.6GHz		41,497.13	
Intel Pentium Dual Core N4200 1.1GHz			57,666.29
Intel Pentium Dual Core 4405Y 1.5GHz		60,946.50	
Intel Pentium Dual Core 4405U 2.1GHz		49,450.90	
Intel Core M m7-6Y75 1.2GHz			1,21,793.90
Intel Core M m3-7Y30 2.2GHz		98,009.90	
Intel Core M M3-6Y30 0.9GHz		66,297.90	
Intel Core M m3 1.2GHz			1,25,103.84
Intel Core M 7Y30 1.0GHz		59,757.30	
Intel Core M 6Y75 1.2GHz	1,94,731.50		1,75,770.37
Intel Core M 6Y54 1.1GHz			1,68,370.90
Intel Core M 6Y30 0.9GHz			72,243.90
Intel Core M 1.2GHz			1,21,100.20
Intel Core M 1.1GHz			1,22,041.65
Intel Core i7 8650U 1.9GHz	1,59,348.84		1,28,631.80
Intel Core i7 8550U 1.8GHz	1,43,592.08		1,09,866.13
Intel Core i7 7Y75 1.3GHz	1,89,842.57		1,56,682.06
Intel Core i7 7820HQ 2.9GHz	2,42,695.90		1,66,851.37
Intel Core i7 7820HK 2.9GHz	2,75,555.81		
Intel Core i7 7700HQ 2.8GHz	1,83,604.32	99,000.90	1,33,720.59
Total	1,76,524.01	55,181.68	1,12,272.94

key insights :

- The tree map compares the relative representation of different groups in the dataset.
- The column chart helps compare values across selected categories.
- The detailed table supports comparison of individual laptop records and their specifications.
- The horizontal bar chart helps rank categories by the selected metric.
- Combining these visuals helps investigate differences between products and categories.



Page 3: Operating System & Specification

- Brand & Specification slicers: Company, RAM/RAM category, Type, OpSys and final_price allow users to filter the report.
- Storage Type Comparison: stacked horizontal bars compare HDD, Hybrid/Other and SSD by company.
- Screen Size vs Price: scatter plot explores screen-size and average-price distribution.
- Weight per Inch Ratio by Brand: ranks brands by the displayed weight-per-inch measure.
- Processor Brand & RAM Split: matrix compares company totals across RAM categories.
- Key insights panel: describes category ranking, grouped comparisons, scatterplot use and detailed specification records.

Interactivity: the slicers on Page 3 let a viewer narrow the visible data by brand, RAM, laptop type, operating system and final-price selection. The cards provide quick summary measures, while charts and tables provide detail.



Operating System & Specification

Brand & Specification

Company

All

Ram, Ram_category

All

Type

All

OpSys

All

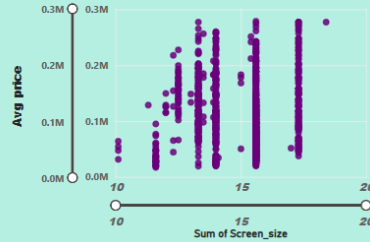
final_price

All

Storage Type Comparison



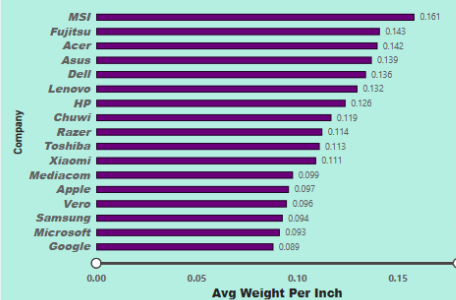
Screen Size vs Price



Key insights:

- The horizontal bar chart supports ranking the displayed categories.
- The clustered bar chart compares multiple groups across categories.
- The scatter plot helps investigate relationships between numerical variables, if those variables are configured on its axes.
- The table provides detailed records for checking individual specifications.

Weight per Inch Ratio by Brand



Processor Brand & RAM Split

Company	2	4	6	8	12	16	24	32	Total
LG				2,08,010.90		2,08,010.90			2,08,010.90
Razer				1,01,973.90		2,17,920.90			1,79,271.90
MSI				1,21,706.06		2,03,134.35		2,77,380.90	1,71,334.80
Google				1,40,424.70		2,17,920.90			1,66,256.77
Microsoft		1,02,964.90		1,65,392.95		2,56,569.90			1,59,779.76
Apple		95,036.90		1,35,728.47		2,35,111.12			1,48,601.29
Huawei				1,41,118.40					1,41,118.40
Samsung		45,982.40		1,58,460.90		1,73,325.90			1,40,072.34
Toshiba		83,449.28		1,26,788.54		1,76,918.28		2,77,380.90	1,25,640.22
Xiaomi				1,03,523.16		1,38,735.05			1,12,326.13
Dell	37,558.90	60,684.14	44,991.40	1,12,304.42	98,288.77	1,74,512.98		2,69,502.45	1,10,725.92
Asus	22,567.55	44,029.58	60,253.30	1,07,265.03	1,10,117.94	1,79,481.18	1,25,757.90	2,02,064.90	1,03,364.36
Lenovo	24,585.72	52,579.94	67,453.33	1,17,208.43	1,36,873.62	1,82,242.09	2,36,056.20	2,63,903.30	1,02,991.52
HP	22,892.10	71,181.75	63,076.49	1,22,152.98	1,62,226.70	1,63,671.91			1,02,121.78
Total	25,208.11	56,940.13	60,966.95	1,15,693.34	1,22,828.35	1,83,702.58	1,80,907.05	2,51,685.69	1,06,047.79

6. Key Insights, Recommendations and Conclusion

Key insights supported by the notebook and dashboard

- Brand and laptop type are useful comparison dimensions: the notebook's grouped charts and dashboard compare brands across multiple product categories.
- The notebook's correlation heatmap records a RAM-to-screen-size correlation of approximately 0.19, indicating a weak positive relationship in the analysed data.
- The notebook's pairplot interpretation says many records fall in the 4–16 GB RAM range and that screen size tends to increase with weight; these are exploratory patterns, not proof of causation.
- The notebook's final summary identifies RAM, CPU, brand, memory and laptop type as important price-related factors. The supplied visuals support comparing these factors, but causal strength has not been established by a predictive model.
- The Power BI dashboard combines KPI cards, ranking charts, scatter plots, treemaps, a matrix/table and slicers so users can explore the dataset from different angles.

Business recommendations

6. Use brand and laptop-type comparisons to plan product assortment, but validate stock recommendations against actual sales and demand data; this dataset describes laptop listings/specifications and does not itself show confirmed units sold or profit.
7. Offer customer comparisons by budget, RAM, storage, screen size, operating system and weight, with the final-price calculation corrected first.
8. Keep premium and budget segments separate when comparing average prices so a few expensive models do not obscure the broader pattern.
9. Use slicers to compare brands and specifications interactively during a presentation, and keep chart titles, units and category labels consistent across pages.
10. Add a short data-quality note to the report describing the IQR filter, derived fields and the exact meaning of the price measure.

Checks recommended before final submission

- Verify the intended price-to-INR conversion and correct the final_price formula; make the Power BI measures match the corrected Python definitions.
- Confirm the KPI aggregation and units: average vs sum, price currency, weight in kilograms, and storage capacity units.
- Check that labels such as “market share”, “revenue” or “performance” are only used where the dataset actually supports them. Laptop listing counts or summed price are not automatically sales, revenue or profit.
- Record the final row count after cleaning and outlier removal, and confirm Power BI is connected to the latest cleaned CSV.
- Check that the values and labels shown in Power BI are consistent with the final cleaned data.

Conclusion

The project follows a complete data-analysis workflow: defining a business question, inspecting a CSV dataset, cleaning and transforming data in Python, performing univariate/bivariate/multivariate EDA, exporting a cleaned dataset, and presenting results in a three-page Power BI dashboard. The dashboard makes laptop price and specification comparisons easier to communicate through summary cards, visual comparisons, detailed tables and filters. Correctly validating the price conversion and final-price calculation is the most important technical check before presenting the dashboard as an India-market pricing report.

Prepared from the uploaded Main_project(1).ipynb notebook and the three supplied Power BI dashboard screenshots.